

Homework 2

MATH 346/377 - Winter 2025

Due on Sunday, February 9th, 9:00pm on Crowdmark

Math 346: Do problems 1 to 7 – Total of points= 85pts

Math 377: Do problems 1 to 9 – Total of points= 110pts

1. (8 points) We say that a positive integer n is a perfect k -th power if $n = m^k$ for some integer m .
 - (a) (4 points) Suppose that a, b are coprime positive integers. Show that the product ab is a perfect k -th power if and only if a and b are both perfect k -th powers.
 - (b) (4 points) Is the statement still correct if a and b are not positive? or if a and b are not coprime? Prove or give a counterexample.

Solution:

- (a) Suppose that a and b are both perfect k -th powers, so $a = x^k$ and $b = y^k$. Then $ab = (xy)^k$ is a perfect power. Conversely, suppose that $ab = n^k$ is a perfect k -th power. Let

$$a = \prod_{p_1|a} p_1^{a_p}, \quad b = \prod_{p_2|b} p_2^{b_p}, \quad n = \prod_{q|n} q^{n_q} \quad (1)$$

be the prime factorizations. Note that the primes dividing a, b are different, since a, b are coprime. It follows that

$$ab = \prod_{p_1|a} p_1^{a_p} \prod_{p_2|b} p_2^{b_p} = \prod_{q|n} q^{kn_q} = n^k. \quad (2)$$

By comparing the prime factorizations, we see that each prime power $p_1^{a_p}$ and $p_2^{b_p}$ on the LHS must be equal to some q^{kn_q} on the RHS, since the primes dividing a and b are distinct. This shows that the exponents a_p and b_p are all divisible by k , so a and b are k -th powers.

- (b) The statement is not correct if a, b are not positive. For example $(-1)(-1) = 1 = 1^2$ is a square, but -1 is not. It is also not correct if a, b are not coprime, for example $3 \cdot 3 = 3^2$ is a square, but 3 is not.

2. (15 points) Are the following statements true or false, where a and b are positive integers and p is prime? In each case, give a proof or a counterexample.
- (a) (3 points) If $\gcd(a, p^2) = p$ then $\gcd(a^2, p^2) = p^2$.
 - (b) (3 points) If $\gcd(a, p^2) = p$ and $\gcd(b, p^2) = p^2$ then $\gcd(ab, p^4) = p^3$.
 - (c) (3 points) If $\gcd(a, p^2) = p$ and $\gcd(b, p^2) = p$ then $\gcd(ab, p^4) = p^2$.
 - (d) (3 points) If $\gcd(a, p^2) = p$ then $\gcd(a + p, p^2) = p$.
 - (e) (3 points) We have $\gcd(a^n, b^n) = \gcd(a, b)^n$.

Solution:

- (a) True. If $\gcd(a, p^2) = p$ then $a = pn$ with n coprime to p , and $a^2 = p^2n^2$ with n^2 coprime to p^2 . Hence $\gcd(a^2, p^2) = p^2$.
- (b) False. Counterexample: if $a = p$ and $b = p^3$ then $ab = p^4$.
- (c) True. If $\gcd(a, p^2) = p$ and $\gcd(b, p^2) = p$, then $a = pn$ and $b = pm$ with m, n coprime to p . So $ab = p^2mn$ with mn coprime to p^2 . Hence $\gcd(ab, p^2) = p^2$.
- (d) False. Take $a = p(p - 1) = p^2 - p$. Then $\gcd(a, p^2) = p$ but $\gcd(a + p, p^2) = \gcd(p^2, p^2) = p^2$.
- (e) Let $d := \gcd(a, b)$. Then $a = dk$ and $b = dl$ with k, l coprime. So $a^n = d^n k^n$ and $b^n = d^n l^n$, and k^n, l^n coprime. This shows that d^n is the gcd of a^n, b^n .

We have seen that there are infinitely many primes. Since 2 is the only even prime, it is immediate that there are infinitely many odd primes i.e. there are infinitely many primes $p \equiv 1 \pmod{2}$. This has been vastly generalized by Dirichlet:

Theorem (Dirichlet's theorem on primes in arithmetic progressions). *Let a and b be two positive coprime integers. There are infinitely many primes $p \equiv a \pmod{b}$ i.e. of the form $p = aq + b$.*

Although it is a difficult theorem to prove in general, the following two exercises prove some special cases.

3. (10 points) Follow Euclid's proof¹ to show that there are infinitely many primes $p \equiv 2 \pmod{3}$.

¹This is the proof we have seen in class: assume there are only finitely many primes $p_1, \dots, p_r$. The integer $p_1 \cdots p_r + 1$ is divisible by a prime that is not in the list, and this is a contradiction.

Solution: Let S be the set of all primes that are congruent to 2 modulo 3. Suppose it is finite and its elements are $S := \{2, p_1, \dots, p_r\}$. The integer $m := 3p_1 \cdots p_r + 2$ is coprime to all primes $\{2, p_1, \dots, p_r\}$. Let

$$m = q_1^{e_1} \cdots q_l^{e_l} \quad (3)$$

be the prime factorization of m . Note that all the primes $q_1, \dots, q_l$ that divide m are not in S , since all the primes in S are coprime to m . By our assumption, for all these primes we have $q_l \equiv 0$ or $1 \pmod{3}$, which also implies that $q_l^{e_l} \equiv 0$ or $1 \pmod{3}$. Taking the product gives

$$m = q_1^{e_1} \cdots q_l^{e_l} \equiv 0 \text{ or } 1 \pmod{3}. \quad (4)$$

This is a contradiction since m is clearly congruent to 2 modulo 3.

Numbers of the form $F_n := 2^{2^n} + 1$ are called Fermat numbers. The only primes of the form $2^m + 1$ must be Fermat numbers, by part a) below. Fermat conjectured that F_n is prime for every $n \geq 0$. This is true for the first few Fermat numbers $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$ and $F_4 = 65537$ which are primes. But in 1732, Euler showed that the next Fermat number

$$F_5 = 2^{2^5} + 1 = 4294967297 = 641 \times 6700417 \quad (5)$$

is not prime. (Observe that the two primes 641 and 6700417 are both congruent to 1 mod 64; see part b) below.) No other prime Fermat number is known.

4. (17 points) The goal of this problem is to use Fermat numbers to show that there are infinitely many primes that are congruent to 1 mod 2^k for any $k \geq 1$.

- (a) (5 points) Show that if $2^m + 1$ is prime, then m is a power of 2.
 (b) (5 points) Show that if a prime p divides the Fermat number F_n , then $p \equiv 1 \pmod{2^{n+1}}$.

Hint: Fermat's little theorem.

- (c) (7 points) Show that distinct Fermat numbers are coprime. For a fixed $k \geq 1$, deduce from b) and c) that there are infinitely many primes that are congruent to 1 mod 2^k .

Hint: First, find a recursion formula for the Fermat numbers, and then a formula for the product $F_0 F_1 \cdots F_n$.

Solution:

- (a) Let us show the contraposition: If m is not a power of 2 then $2^m + 1$ is not prime. We write $m = 2^k q$ with q coprime to 2 (so q is odd). We have seen in the lectures that if q is odd, then the polynomial $x^q + 1$ is divisible by $x + 1$ since

$$\frac{x^q + 1}{x + 1} = \frac{1 - (-x)^q}{1 - (-x)} = 1 - x + x^2 + \cdots + (-x)^{q-1}. \quad (6)$$

If we evaluate at $x = 2^{2^k}$ we find

$$2^{2^k q} + 1 = (2^{2^k} + 1)(1 - 2^{2^k} + 2^{2 \cdot 2^k} + \cdots + 2^{2^k(q-1)}) \quad (7)$$

and $2^{2^k q} + 1$ is divisible by $2^{2^k} + 1$.

- (b) If p divides $F_n = 2^{2^n} + 1$, then $2^{2^n} \equiv -1 \pmod{p}$ and $(2^{2^n})^2 = 2^{2^{n+1}} \equiv 1 \pmod{p}$. This shows that the order of 2 modulo p is a divisor of 2^{n+1} . It has to be of the form 2^r for some $0 \leq r \leq n + 1$. Assume that $r \leq n$. Then we would have

$$2^{2^r} \equiv 1 \pmod{p} \quad (8)$$

and

$$1 \equiv 1^{2^{n-r}} \equiv (2^{2^r})^{2^{n-r}} \equiv 2^{2^{n-r} 2^r} \equiv 2^{2^n} \equiv -1 \pmod{p}. \quad (9)$$

We conclude that the order of 2 is 2^{n+1} . On the other hand, since 2 is coprime to p , Fermat's little theorem implies that $2^{p-1} \equiv 1 \pmod{p}$. That shows that the order of 2 divides $p - 1$, in other words

$$p \equiv 1 \pmod{2^{n+1}}. \quad (10)$$

- (c) We have

$$F_{n+1} = 2^{2^{n+1}} + 1 = 2^{2^{n+1}} + 1 = 2^{2^n} 2^{2^n} + 1 \quad (11)$$

$$= (2^{2^n} + 1)2^{2^n} - (2^{2^n} + 1) + 2 \quad (12)$$

$$= (2^{2^n} - 1)F_n + 2. \quad (13)$$

So $F_{n+1} = a_n F_n + 2$ with $a_n := 2^{2^n} - 1$. On the other hand, we also have

$$a_n = 2^{2^n} - 1 = 2^{2^{n-1}} 2^{2^{n-1}} - 1 = (2^{2^{n-1}} - 1)2^{2^{n-1}} + 2^{2^{n-1}} - 1 = a_{n-1} F_{n-1}. \quad (14)$$

So we find that $F_{n+1} = a_{n-1} F_{n-1} F_n + 2$, and by iterating we find that

$$F_{n+1} = a_0 F_0 F_1 \cdots F_n + 2 = F_0 F_1 \cdots F_n + 2. \quad (15)$$

Note that $a_0 = 1$. This shows that for any $k > 0$ we have that $\gcd(F_n, F_{n+k})$ divides 2, and since the Fermat numbers are odd, we deduce that they are all coprime.

Finally, let us fix some $k \geq 1$. There are infinitely many Fermat numbers F_n with $k - 1 \leq n$, all divisible by at least one prime, and all the prime divisors must be distinct. Moreover, any prime dividing F_n must be congruent to 1 modulo 2^{n+1} , so also congruent to 1 modulo 2^k (since $k \leq 2^{n+1}$). It follows that there are infinitely many primes $p \equiv 1 \pmod{2^k}$.

5. (12 points) Let $a, b \geq 2$ be coprime integers, and $n \geq 3$ an even integer. Show that $a + b$ divides $a^{n+1} + b^{n+1}$ and that

$$\gcd\left(a + b, \frac{a^{n+1} + b^{n+1}}{a + b}\right) \text{ divides } n + 1. \quad (16)$$

Hint: Consider the polynomial division of $\frac{x^{n+1} + b^{n+1}}{x + b}$ by $x + b$.

Solution: Let $p(x)$ be the polynomial $\frac{x^{n+1} + b^{n+1}}{x + b}$. We have

$$p(x) = x^n - bx^{n-1} + b^2x^{n-2} + \cdots - b^{n-1}x + b^n \in \mathbb{Z}[x]. \quad (17)$$

Hence $p(a)$ is an integer and we have $a^{n+1} + b^{n+1} = p(a)(a + b)$. This shows that $a + b$ divides $a^{n+1} + b^{n+1}$.

Now let $h(x) \in \mathbb{Z}[x]$ be the polynomial of degree $n - 1$, and $r \in \mathbb{Z}$ be such that $p(x) = (x + b)h(x) + r$. We write $h(x) = \sum_{l=0}^{n-1} c_l x^l$ so that

$$\sum_{k=0}^n (-1)^k b^{n-k} x^k = p(x) = (x + b) \left(\sum_{l=0}^{n-1} c_l x^l \right) + r \quad (18)$$

$$= \sum_{l=1}^n c_{l-1} x^l + \sum_{l=0}^{n-1} b c_l x^l + r \quad (19)$$

$$= c_{n-1} x^n + \left(\sum_{k=1}^{n-1} (c_{k-1} + b c_k) x^k \right) + b c_0 + r. \quad (20)$$

Comparing the coefficients gives

$$c_{n-1} = (-1)^n = 1, \quad (21)$$

$$c_{k-1} + b c_k = (-1)^k b^{n-k} \quad \text{for } 1 \leq k \leq n - 1, \quad (22)$$

$$b c_0 + r = b^n. \quad (23)$$

We find that $c_{n-2} = -2b$, $c_{n-3} = 3b^2, \dots$ and more generally $c_{n-k} = k(-b)^{k-1}$. Hence, we have $c_0 = -nb^{n-1}$

$$r = b^n - bc_0 = (n+1)b^n. \quad (24)$$

Note that $h(x) \in \mathbb{Z}[x]$, so that $h(a) \in \mathbb{Z}$. From the polynomial division we deduce that at $x = a$ we have

$$\frac{a^{n+1} + b^{n+1}}{a+b} = p(a) = (a+b)h(a) + (n+1)b^n. \quad (25)$$

Hence, if d divides $p(a)$ and $a+b$, it must divide $(n+1)b^n$. Now any prime that divides d must be coprime to b . Otherwise it would divide a since it divides $a+b$. Hence d is coprime to b and divides $n+1$.

6. (11 points) Let p be a prime. Show that $\sqrt{2} + \sqrt{p+2}$ is always an irrational number.

Solution: There are different ways to solve this. The fastest is to suppose that $\sqrt{2} + \sqrt{p} = m/n$ is a rational number, and by squaring both sides one finds the contradiction that $\sqrt{2}$ is rational.

On the other hand, one can use the (more general method) that uses the rational root theorem, as follows. Let $\alpha = \sqrt{2} + \sqrt{p}$. Then

$$\alpha^2 = (\sqrt{2} + \sqrt{p+2})^2 = p+4 + 2\sqrt{2(p+2)} \quad (26)$$

and

$$\alpha^4 = (p+4 + 2\sqrt{2(p+2)})^2 = (p+4)^2 + 4(p+4)\sqrt{2(p+2)} + 8(p+2). \quad (27)$$

If we substitute $2\sqrt{2(p+2)} = \alpha^2 - (p+4)$ we find that

$$\alpha^4 = 2(p+4)\alpha^2 + 8(p+2) - (p+4)^2 = 2(p+2)\alpha^2 - p^2. \quad (28)$$

This implies that α is a root of the monic polynomial

$$f(x) = x^4 - 2(p+4)x^2 + p^2. \quad (29)$$

By the rational root theorem, any rational root must be an integer dividing p^2 so the possible rational roots are $\pm 1, \pm p \pm p^2$.

If $x = \pm 1$ was a root, we would have

$$p^2 - 2p - 7 = 0. \quad (30)$$

If $p \neq 7$ we can reduce modulo p to find the contraction $0 \equiv 7 \pmod{p}$. If $p = 7$ then $7^2 - 2 \cdot 7 - 7 = 28 \neq 0$.

If $x = \pm p$ was a root, we would have $p^4 - 2(p+4)p^2 + p^2 = 0$ which also implies

$$p^2 - 2p - 7 = 0. \quad (31)$$

If $x = \pm p^2$ was a root, we would have $p^8 - 2(p+4)p^4 + p^2 = 0$ which would imply that

$$p^6 - 2p^3 - 8p^2 + 1 = 0. \quad (32)$$

Reducing modulo p shows again that this cannot hold.

Hence, $f(x)$ has no rational roots and this shows that $\sqrt{2} + \sqrt{p+2}$ is irrational.

Let $\pi(x)$ be the number of primes $p \leq x$. So for example $\pi(1) = 0$, $\pi(2) = 1$, $\pi(10) = 4$ and $\pi(20) = 8$ (the primes less than 20 are 2, 3, 5, 7, 11, 13, 17, 19). In class I have mentioned the Prime Number Theorem, that states that

$$\lim_{n \rightarrow \infty} \frac{\pi(n) \log(n)}{n} = 1. \quad (33)$$

It is known that there are always at least a prime between n and $2n$, and the PNT implies that the density goes to 0 as $n \rightarrow \infty$. More precisely, we have

$$\lim_{n \rightarrow \infty} \frac{|\{\text{primes } p, n < p \leq 2n\}|}{n} = \lim_{n \rightarrow \infty} \frac{\pi(2n) - \pi(n)}{n} = 0. \quad (34)$$

The following exercise proves this fact without using the PNT.

7. (12 points) Show that if p is a prime with $n < p \leq 2n$ then p divides the binomial coefficient $\binom{2n}{n}$. Deduce that

$$n^{\pi(2n) - \pi(n)} < 2^{2n}, \quad (35)$$

and hence $\frac{\pi(2n) - \pi(n)}{n} < 2 \log(2) / \log(n)$. (This shows that the limit goes to 0 as $n \rightarrow \infty$, hence the density of primes between n and $2n$ goes to 0).

Solution: The binomial coefficient

$$\binom{2n}{n} = \frac{(2n)!}{n!n!} = \frac{(n+1) \cdots (2n-1)(2n)}{1 \cdot 2 \cdots n}. \quad (36)$$

is an integer. Let $a := (n+1) \cdots (2n-1)(2n)$ and $b := 1 \cdot 2 \cdots n$, so b divides a and $\binom{2n}{n} = \frac{a}{b}$. If p is a prime $n < p \leq 2n$, then it must appear in the product a , which means that p divides a . We write $a = pa'$. On the other hand p is coprime to b , since p is prime not appearing in the product $1 \cdots n$ (we assumed $n < p$). Hence b divides $a = pa'$ and is coprime to p , by Euclid's Lemma this means that b divides a' . We write $a' = bk$ for some $k \in \mathbb{Z}$, and it follows that

$$\binom{2n}{n} = \frac{a}{b} = \frac{pbk}{b} = pk. \quad (37)$$

This shows that any prime $n < p \leq 2n$ divides $\binom{2n}{n}$. For the second part, note that

$$2^{2n} = (1+1)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} > \binom{2n}{n}. \quad (38)$$

Let us denote by $p_1, \dots, p_r$ the prime numbers between n and $2n$. So by definition, we have $r := \pi(2n) - \pi(n)$. Since each p_i divides $\binom{2n}{n}$ but the p_i 's are coprime, it follows that $p_1 \cdots p_r$ divides $\binom{2n}{n}$. Finally, since all the primes are greater than n we deduce that

$$n^{\pi(2n)-\pi(n)} = n^r < p_1 \cdots p_r < \binom{2n}{n} < 2^{2n}. \quad (39)$$

Taking the logarithm on both sides gives $\frac{\pi(2n)-\pi(n)}{n} < 2 \log(2) / \log(n)$

Problems 8 and 9 for MATH 377 students.

We have seen in class that “there are more primes than squares”, which can be deduced from the PNT. This can also be seen from the divergence of

$$\lim_{N \rightarrow \infty} \sum_{i=1}^N \frac{1}{p_i} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} \cdots = \infty \quad (40)$$

where p_i is the i -th prime number, since on the other hand the sum $\sum_{k=1}^{\infty} 1/k^2$ converges to $\pi/6$ (You can assume this fact without proof. Note also that $\sum_{k=1}^{\infty} 1/k$ diverges). The following exercise is a proof of the divergence of (40).

8. (12 points) An integer $n > 1$ is said to be square-free² if any prime appearing in its factorization appears with exponent 1. Show that any integer $n > 1$ can be written $n = k^2m$ where m is square-free, and that we have

$$\lim_{N \rightarrow \infty} \sum_{\substack{1 \leq m \leq N \\ m \text{ squarefree}}} \frac{1}{m} = \infty. \quad (41)$$

Deduce that

$$\lim_{N \rightarrow \infty} \prod_{i=1}^N \left(1 + \frac{1}{p_i}\right) = \infty, \quad (42)$$

and conclude the divergence of (40).

Hint: We have $1 + x \leq e^x$.

Solution: Let $n = \prod_p p^{a_p}$ be the prime factorization of some integer n . For each p we can write $a_p = 2q_p + r_p$ with $r_p = 0$ or 1 . Hence we have

$$n = \prod_p p^{a_p} = \prod_p p^{2q_p + r_p} = \left(\prod_p p^{q_p}\right)^2 \left(\prod_p p^{r_p}\right) = k^2 n, \quad (43)$$

where $n = \prod_p p^{r_p}$ is squarefree.

Let $S(N)$ be the sum

$$S(N) := \sum_{\substack{1 \leq m \leq N \\ m \text{ squarefree}}} \frac{1}{m}. \quad (44)$$

and $H(N)$ the sum

$$H(N) := \sum_{n=1}^N \frac{1}{n}. \quad (45)$$

We can write any n as $n = k^2m$, so that

$$H(N) = \sum_{n=1}^N \frac{1}{n} \leq \sum_{1 \leq k \leq \sqrt{N}} \sum_{\substack{1 \leq m \leq N \\ m \text{ squarefree}}} \frac{1}{k^2 m} \quad (46)$$

$$= \left(\sum_{1 \leq k \leq \sqrt{N}} \frac{1}{k^2} \right) \left(\sum_{\substack{1 \leq m \leq N \\ m \text{ squarefree}}} \frac{1}{m} \right) \quad (47)$$

$$= \left(\sum_{1 \leq k \leq \sqrt{N}} \frac{1}{k^2} \right) S(N). \quad (48)$$

²The integer $105 = 3 \cdot 5 \cdot 7$ is square-free, but $54 = 2 \cdot 3^3$ is not.

As $N \rightarrow \infty$, the harmonic sum $H(N)$ diverges, and so does $S(N)$. (Note that the sum of $1/k^2$ in front converges to $\pi/6$, but this is not important here).

Let us denote by $P(N)$ the product

$$P(N) = \prod_{i=1}^N \left(1 + \frac{1}{p_i}\right) \quad (49)$$

$$= 1 + \sum_{i=1}^N \frac{1}{p_i} + \sum_{\substack{1 \leq i_1, i_2 \leq N \\ i_1 \neq i_2}} \frac{1}{p_{i_1} p_{i_2}} + \cdots. \quad (50)$$

We see that we get the sum of $1/m$ where m are all the possible square free integers between 1 and $p_1 \cdots p_N$. Thus, we have

$$P(N) = S(p_1 \cdots p_N). \quad (51)$$

Taking the limits give

$$\lim_{N \rightarrow \infty} P(N) = \lim_{N \rightarrow \infty} S(p_1 \cdots p_N) = \infty, \quad (52)$$

since $\lim_{N \rightarrow \infty} p_1 \cdots p_N = \infty$ and S diverges. Finally, using the hint we get that

$$P(N) = \prod_{i=1}^N \left(1 + \frac{1}{p_i}\right) \leq \prod_{i=1}^N \exp\left(\frac{1}{p_i}\right) = \exp\left(\sum_{i=1}^N \frac{1}{p_i}\right). \quad (53)$$

We have seen that we can use factorization in ring of integers to solve diophantine equations, for example the equation $2x^3 = y^2 + 1$. It relied on the following proposition, which is an immediate generalization of Exercise 2. (The proof is done in the exact same way by using the unique factorization into irreducibles, but you can use it without proof in the next problem.)

Proposition. *Let R be a Euclidean domain and suppose that we have $\alpha_1 \alpha_2 = \gamma^k$ for some nonzero $\alpha_1, \alpha_2, \gamma \in R$. If α_1 and α_2 are coprime, then there exist elements $\gamma_1, \gamma_2 \in R$ and units³ $u_1, u_2 \in R^\times$ such that $\alpha_i = u_i \gamma_i^k$.*

9. (13 points) The ring $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain with Euclidean function $|a + \sqrt{-2}ib|^2 = a^2 + 2b^2$. (You can assume it without proof). We want to find all integers $x, y \in \mathbb{Z}$ satisfying the equation $x^3 = y^2 + 2$.

³In class I only considered the case where the unit is 1 because in $\mathbb{Z}[i]$ any unit is a cube ($1^3 = 1, (-1)^3 = -1, i^3 = -i$ and $(-i)^3 = i$). The same happens in the exercise where you can also ignore the unit)

- (a) (2 points) Show that the group of units is $\mathbb{Z}[\sqrt{-2}]^\times = \{\pm 1\}$ and that $i\sqrt{2}$ is irreducible.
- (b) (4 points) Reduce the equation modulo 4 to deduce that x and y are both odd.
- (c) (7 points) Following the example that we have seen in class solve the equation $x^3 = y^2 + 2$.
Hints: You should find two solutions. You can use the above proposition without proof.

Solution:

- (a) An element $a + b\sqrt{-2}$ is a unit if and only if $a^2 + 2b^2 = 1$. This implies $b = 0$ and $a = \pm 1$. The element $i\sqrt{2}$ is irreducible since it has prime norm $|i\sqrt{2}|^2 = 2$.
- (b) If y is even, then $y^2 \equiv 0 \pmod{4}$, so that we would have $x^3 \equiv 2 \pmod{4}$. But there is no such residue class $x \pmod{4}$, and y has to be odd. This implies that $y^2 \equiv 1 \pmod{4}$, hence $x^3 \equiv 3 \pmod{4}$. It follows that $x \equiv 3 \pmod{4}$, so x is odd as well.
- (c) We factor in $\mathbb{Z}[\sqrt{2}]$ as $x^3 = (y + i\sqrt{2})(y - i\sqrt{2})$. We have

$$(y + i\sqrt{2}) - (y - i\sqrt{2}) = 2i\sqrt{2} = -(i\sqrt{2})^3. \quad (54)$$

This implies that the gcd of $y + i\sqrt{2}$ and $y - i\sqrt{2}$ is $\pm(i\sqrt{2})^k$ for some $k \in \{0, 1, 2, 3\}$. Since $y = 2k + 1$ is odd we write

$$y + i\sqrt{2} = (2k + 1) + i\sqrt{2} = -(i\sqrt{2})^2 k + i\sqrt{2} + 1 = i\sqrt{2}(1 - i\sqrt{2}k) + 1. \quad (55)$$

Since we can write

$$(y + i\sqrt{2}) - i\sqrt{2}(1 - i\sqrt{2}k) = 1 \quad (56)$$

it follows that $y + i\sqrt{2}$ and $i\sqrt{2}$ are coprime. Hence, we deduce that $y + i\sqrt{2}$ and $y - i\sqrt{2}$ are coprime, and from the proposition above we find that

$$y + i\sqrt{2} = (m + ni\sqrt{2})^3 = (m^3 - 6mn^2) + (3m^2n - 2n^3)i\sqrt{2}. \quad (57)$$

for some $m, n \in \mathbb{Z}$. Comparing coefficients shows that

$$1 = 3m^2n - 2n^3 = n(3m^2 - 2n^2) \quad (58)$$

$$y = m^3 - 6mn^2 = m(m^2 - 6n^2). \quad (59)$$

From the first equation we deduce that $n = 3m^2 - 2n^2 = \pm 1$. If $n = -1$, then $-1 = 3m^2 - 2n^2 = -3m^2 - 2$ implies that $1 = -3m^2$ which is impossible. So we deduce that $n = 1$ and $m^2 = 1$. Substitution of $(m, n) = (\pm 1, 1)$ in the second equation gives $y = \pm 5$. Finally, this implies that

$$x^3 = (\pm 5)^2 + 2 = 27 = 3^3, \quad (60)$$

and we conclude that the two solutions are $(x, y) = (3, 5)$ and $(x, y) = (3, -5)$.